

**Score: 2/5 — Significant gaps** | **Confidence: high**

**Benchmark:** ARABICMMLU | **Context:** Non-Arab Tourists and Expats in Eight Arabic-Speaking Countries (ArabicMMLU Deployment)

**Output Content definition:** Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

Label-correctness verification process — 96% inter-annotator agreement on 100 sampled questions [Q42, Q43] with native MSA speakers holding Bachelor's degrees and competitive compensation [Q35, Q36, Q39] — establishes a reasonable practical ceiling [Q44]. However, annotator-pool composition undermines convergent validity for the multi-country deployment: 10 annotators are from Jordan (4), Egypt (2), Lebanon (1), UAE (1), KSA (2), with no Moroccan, Kuwaiti, or Palestinian annotators [Q31] despite Palestine being a top-3 source country [WEB-3, WEB-4]. For contested historical questions where 'correct' answers vary across national curricula, the 96% agreement likely reflects Jordanian/Egyptian consensus rather than pan-regional validity. The sampled data shows concrete examples of Jordanian-curriculum framing for shared Arab history [DATASET-D1, DATASET-D2, DATASET-D3, DATASET-D11]. Training data contamination is acknowledged as unverifiable [Q95]. OCR corruption in university Egyptian items [DATASET-D23, DATASET-D24] suggests label quality may degrade in some sub-corpora. No aggregation method considerations for minority perspectives are documented.

| ID  | Evidence                                                                                                                                                                                                                                                                                                                                                       |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q31 | "The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)                                                                                                        |
| Q35 | "Our internal workers are Master's students and Research Assistants in Computer Science, while the external workers hold Bachelor's degrees." (p.4)                                                                                                                                                                                                            |
| Q36 | "We ensured competitive compensation for the workers, exceeding the monthly average wage in each respective country." (p.4)                                                                                                                                                                                                                                    |
| Q39 | "While our workers are native speakers of Modern Standard Arabic with at least Bachelor's degrees, we maintain the quality of our dataset construction through meticulous steps." (p.4)                                                                                                                                                                        |
| Q42 | "Finally, we assessed the accuracy of our data collection by having two native Arabic speakers annotate 100 randomly sampled questions. They were provided with all metadata, including the answer key, and tasked with verifying the correctness of each sample using any available resources (e.g., search engines)." (p.4)                                  |
| Q43 | "We found that 96% of the questions and answer keys match on average, while the remaining could have incorrect answer keys." (p.4)                                                                                                                                                                                                                             |
| Q44 | "This 96% score is considered to represent the maximum score meaningfully achievable for ArabicMMLU." (p.4)                                                                                                                                                                                                                                                    |
| Q95 | "to a lack of sufficient information about its training regimen. As such, we cannot assert that the model's pretraining data is free from contamination." (p.10)                                                                                                                                                                                               |
| D1  | دحو مامت: ني طسلف يف اعامطأو ةي نو يوصل ة ك رحل نم لولأا هلل ادبع كل مل الم فق اوم نم تسيل ةي تال نم ة دحاو هضفر / ني طسلف ة مال قودنص مدع يل برعلا ةودع / قاربلا طئاح يل ارمم دوهيلا اعطاع هضفر / م 1950 ماع ني تفضلا<br>Jordan HS history question about King Abdullah I's positions on Zionism and Palestine — framed from Jordanian curriculum perspective |
| D2  | نب لصل يف رم 1-2 / ني سحلا نب هلل ادبع رم 1-2: سي راب يف ءافلحل ال دقع يذل اخلصل رمتؤم يف برعلا _____ لثم<br>Jordan civics HS: who represented Arabs at Paris Peace Conference — Jordanian Hashemite framing                                                                                                                                                   |
| D3  | Jordan civics HS: — بادتالال كف دويقي نم دالبل ررح: 1928 ماع هي نا طي ربل هي ندرألا ده اعملل سي رئل فدهلناك<br>primary goal of the 1948 Jordanian-British treaty — Jordanian national framing                                                                                                                                                                  |
| D11 | Jordan primary social science: fact about King Talal's reign — Jordan-specific<br>في كل ذو ةي حص باب سأل هتالو تهتنا ني سحلا نب لال كل مل                                                                                                                                                                                                                      |
| D23 | Severely garbled OCR text making question almost illegible<br>ة زكرم لا: لي ري تذت... صاخا لال ني تب ة طل تدلا عتي هت ة تجرد                                                                                                                                                                                                                                   |

|              |                                                                                                                                                                                                                  |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>D24</b>   | قـبـس ام لك؟ مل اـرـسـالـا رـلـوـدـل اـرـف مـكـاـر لـا رـف اـر رـفـاـوت بـرر رـتـل اـرـسـالـا طـوـرـرـشـل اـنـرـم — Egypt university political science with severe OCR corruption — content partially unreadable |
| <b>WEB-3</b> | Palestine confirmed top-3 source country — yet no Palestinian annotators in pool ( <a href="https://github.com/mbzuai-nlp/ArabicMMLU">https://github.com/mbzuai-nlp/ArabicMMLU</a> )                             |
| <b>WEB-4</b> | HuggingFace dataset card confirms Palestine major contributor ( <a href="https://huggingface.co/datasets/MBZUAI/ArabicMMLU">https://huggingface.co/datasets/MBZUAI/ArabicMMLU</a> )                              |

## Strengths

- 96% answer-key accuracy verification on a 100-question sample provides a documented quality ceiling [Q42, Q43, Q44]
- Native-speaker annotator workforce with at least Bachelor's degrees and competitive pay [Q35, Q36, Q39]
- Pre-collection workshop standardized annotator process [Q40]
- Annotators include workers from 5 of 8 deployment countries (Jordan, Egypt, Lebanon, UAE, KSA) [Q31, Q96]

| ID         | Evidence                                                                                                                                                                                                                                                                                                                      |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q31</b> | "The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)                                                                       |
| <b>Q35</b> | "Our internal workers are Master's students and Research Assistants in Computer Science, while the external workers hold Bachelor's degrees." (p.4)                                                                                                                                                                           |
| <b>Q36</b> | "We ensured competitive compensation for the workers, exceeding the monthly average wage in each respective country." (p.4)                                                                                                                                                                                                   |
| <b>Q39</b> | "While our workers are native speakers of Modern Standard Arabic with at least Bachelor's degrees, we maintain the quality of our dataset construction through meticulous steps." (p.4)                                                                                                                                       |
| <b>Q40</b> | "Firstly, we conducted a 1-hour workshop before the data collection stage to clarify the process." (p.4)                                                                                                                                                                                                                      |
| <b>Q42</b> | "Finally, we assessed the accuracy of our data collection by having two native Arabic speakers annotate 100 randomly sampled questions. They were provided with all metadata, including the answer key, and tasked with verifying the correctness of each sample using any available resources (e.g., search engines)." (p.4) |
| <b>Q43</b> | "We found that 96% of the questions and answer keys match on average, while the remaining could have incorrect answer keys." (p.4)                                                                                                                                                                                            |
| <b>Q44</b> | "This 96% score is considered to represent the maximum score meaningfully achievable for ArabicMMLU." (p.4)                                                                                                                                                                                                                   |
| <b>Q96</b> | "We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)                                                                                                                                                                     |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                               |
|-------------|-------------------------------------------------------------------|
| <b>OC-1</b> | Do ground-truth labels reflect regional stakeholder perspectives? |
| <b>OC-2</b> | Assess potential annotator-regional population disagreement       |
| <b>OC-3</b> | Review annotator demographics from documentation                  |
| <b>OC-4</b> | Consider label re-annotation by regional annotators               |
| <b>OC-5</b> | Review aggregation methods for minority perspective erasure       |
| <b>OC-6</b> | Document label issues harming convergent/external validity        |

### Checklist responses:

**OC-1:** Partially — labels reflect curriculum-stated 'correct' answers from source exams, which encode national perspectives. For non-contested STEM content this is acceptable [DATASET-D21, DATASET-D38]; for contested civic/historical content [DATASET-D1, DATASET-D3], labels reflect source-country curriculum perspective without acknowledgment.

**OC-2:** High potential for disagreement on contested content — Palestinian, Moroccan, and Kuwaiti perspectives are not represented in the verifier pool [Q31] despite Palestine being a major source [WEB-3, WEB-4]. The 96% figure [Q43] likely overstates regional validity for politically sensitive items.

**OC-3:** Documented: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 UAE, 2 KSA), 4 external workers (3 Jordanian, 1 Egyptian) [Q31]; internal workers are Master's/RA-level CS students, external workers hold Bachelor's [Q35]. Demographics are documented but skew Jordanian/Egyptian.

**OC-4:** Re-annotation by Palestinian, Moroccan, and Kuwaiti annotators recommended for contested-content items, especially history/civics from those source countries.

**OC-5:** No aggregation method for minority/competing perspectives is documented — the benchmark's structure precludes it [Q27].

**OC-6:** Documented: (1) annotator-pool skew toward Jordan/Egypt with absent Moroccan/Palestinian/Kuwaiti perspectives [Q31, Q96]; (2) 96% ceiling does not account for cross-curriculum disagreement on contested content [Q43, Q44]; (3) OCR corruption in some sub-corpora [DATASET-D23, DATASET-D24]; (4) contamination unverifiable [Q95].

| ID    | Evidence                                                                                                                                                                                                                                                                                                                                           |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q27   | "Each question has 2–5 candidate answers, with one correct answer." (p.3)                                                                                                                                                                                                                                                                          |
| Q31   | "The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)                                                                                            |
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| Q95   | "to a lack of sufficient information about its training regimen. As such, we cannot assert that the model's pretraining data is free from contamination." (p.10)                                                                                                                                                                                   |
| Q96   | "We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)                                                                                                                                                                                          |
| D1    | دحو مامت؛ نيٽسلف ي ف اهعاطأو ؤي نوي هصل اءكءرل نم لولأل هل ادبع كللم الفقاوم نم تسيل ؤي تالآ نم دءاو<br>هضفر / نيٽسل فل اءمأل قودنص معد يل برعل اوءد / قارب طئاح يل ارم دوھي اعطاع هضفر / م ۱۹۵۰ اع نيٽفضل<br>Jordan HS history question about King Abdullah I's positions on Zionism and Palestine — framed from Jordanian curriculum perspective |
| D3    | Jordan civics HS: بادنالك كف دويق نم دالب رحت : ۱۹۲۸ مع هي نايطربل هي ندراأل هدهاعم لل سيئرل فدهلناك<br>primary goal of the 1948 Jordanian-British treaty — Jordanian national framing                                                                                                                                                             |
| D21   | تا دي بل لل عي نصنع ؤل وئسم ؤي لاتل تا يضعل اءىء / X100 وه ئي واصل ره جمليف ؤي تزل سءعل ري بك را دم<br>Palestine HS biology: standard cellular biology — culturally neutral STEM<br>أساسل لم ؤي مزال بو دنالا ءكبشل: ؤي خل ليف                                                                                                                     |
| D23   | Severely garbled OCR text making question almost illegible — ؤزكرم لا: يل ري تذت... صاخا تلأل ني تب ؤتل تدل عتي هي ته ؤءجرد                                                                                                                                                                                                                        |
| D24   | Egypt university political science with severe OCR corruption — قبس ام لك: لم ارسل ال رل ودل اررف مك ارل اررف ار رف او ت برر رر تل ارسل ال طور رر شل ررم<br>content partially unreadable                                                                                                                                                           |
| D38   | + موي سينغامال؟ لصح يذل لعافلت فصت ؤي لاتل اتل داعمل أ: رب خمل اب موي سينغامال نم ؤيم كقرب دءأاق<br>Natural science MS: chemistry reactions — country-neutral STEM<br>— موي سينغامال ديسك -> ني جسكأ                                                                                                                                               |
| WEB-3 | Palestine confirmed top-3 source country — yet no Palestinian annotators in pool ( <a href="https://github.com/mbzuai-nlp/ArabicMMLU">https://github.com/mbzuai-nlp/ArabicMMLU</a> )                                                                                                                                                               |
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## Information Gaps

- Whether the 100-question verification sample included contested-content items
- Distribution of OCR corruption across the full dataset

## Requires Expert Verification

- Whether Palestinian, Moroccan, and Kuwaiti subject-matter experts would agree with ground-truth labels on country-sourced contested items

## Remediation

**Gap 1: Annotator pool lacks Palestinian, Moroccan, and Kuwaiti members despite Palestine being a top-3 source country [Q31, WEB-3]**

**Recommendation:** Re-verify a stratified sample of Palestinian-, Moroccan-, and Egyptian-sourced contested-content items with annotators from those countries; report disagreement rates separately from the 96% pooled figure to expose curriculum-specific disagreement.

| ID    | Evidence                                                                                                                                                                                                                                                |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q31   | "The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4) |
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